Finance 513 Spring 2026
Advanced Financial Derivatives Martin Widdicks

Problem Set 4
Total 15 points
Due Monday 7th April 2025 before 11:59pm

Instructions: This is an **individual assignment as you all need to practice this**! Handwritten solutions are acceptable here, but please submit a scanned or photographed copy to Canvas. Of course, you can also typeset your solutions (word, latex etc.) them and upload them electronically.

Please make sure that your solutions are well-organized and clear, with appropriate text, explanations, and formatting. If I cannot figure out what you did, then what you did is wrong.

Question 1: (2 points total) Practicing using Ito's lemma and the Ito integral, (X is a Brownian motion in the questions below)

a) By considering the process followed by *f(X(t))* = $X^2(t)$, please show that

$$\int_0^t X(\tau)dX(\tau) = \frac{1}{2}X^2(t) - \frac{t}{2}$$

Hint: Use Itô's lemma to compute *df* of when $f(X(t)) = X^2(t)$, and then rearrange.

***Solution*: Applying Itô's formula to $f(t) = X^2(t)$:**

$$\mathrm{d}f(t) = \left(\frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial X}\right)\mathrm{d}t + \frac{\partial f}{\partial X}\mathrm{d}X$$

$$\Rightarrow \mathrm{d}\left(X(t)^2\right) = \mathrm{d}t + 2X(t)\mathrm{d}X(t)$$

$$\Leftrightarrow \mathrm{d}\left(X^2(t)/2\right) = (1/2)\mathrm{d}t + X(t)\mathrm{d}X(t)$$

Integrating both sides, recalling that $X(0) = 0$, and re-arranging the terms we find:

$$\int_0^t d\left(\frac{1}{2}X^2(\tau)\right) = \frac{1}{2}\int_0^t d\tau + \int_0^t X(\tau)dX(\tau)$$

$$\Rightarrow \int_0^t X(\tau)dX(\tau) = \frac{1}{2}\left(X^2(t) - t\right)$$

b) Use a similar approach to show that

$$\int_0^t \tau dX(\tau) = tX(t) - \int_0^t X(\tau)d\tau$$

Solution**: Choose $f = tX$. Applying Itô's formula,**

$$\mathrm{d}f(t) = \left(\frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial X^2}\right)\mathrm{d}t + \frac{\partial f}{\partial X}\mathrm{d}X(t)$$
$$\Rightarrow \mathrm{d}\big(tX(t)\big) = X(t)dt + t\mathrm{d}X(t).$$

Integrating both sides, recalling that $X(0) = 0$, and re-arranging the terms we find

$$\int_0^t \mathrm{d}\big(\tau X(\tau)\big) = \int_0^t X(\tau)\mathrm{d}\tau + \int_0^t \tau\mathrm{d}X(\tau)$$
$$\Rightarrow \int_0^t \tau\mathrm{d}X(\tau) = tX(t) - \int_0^t X(\tau)\mathrm{d}\tau$$

c) and

$$\int_0^t X^2(\tau)dX(\tau) = \frac{1}{3}X^3(t) - \int_0^t X(\tau)d\tau$$

Solution**: Choose $f = (1/3)X(t)^3$. Applying Itô's lemma:**

$$\mathrm{d}f(t) = \left(\frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial X^2}\right)\mathrm{d}t + \frac{\partial f}{\partial X}\mathrm{d}X(t)$$
$$\Rightarrow \mathrm{d}\left(\frac{1}{3}X(t)^3\right) = X(t)\mathrm{d}t + X(t)^2\mathrm{d}X(t).$$

Integrating both sides, recalling that $X(0) = 0$, and re-arranging terms, we find:

$$\int_0^t \mathrm{d}\left(\frac{1}{2}X(\tau)^3\right) = \int_0^t X(\tau)\mathrm{d}(\tau) + \int_0^t X(\tau)^2\mathrm{d}X(\tau)$$
$$\Rightarrow \int_0^t = X(\tau)^2 dX(\tau) = \frac{1}{2}X(t)^3 - \int_0^t X(\tau)\mathrm{d}\tau.$$

<u>Question 2 (2 points): Process followed by the Futures (or Forward) Price.</u>

Let X be a Brownian motion. The price of a dividend paying stock follows the process

$$dS(t) = (\mu - \delta)S(t)dt + \sigma S dX(t)$$

with initial condition S(0) = S_0, where μ is the expected rate of return, δ is the dividend yield, and $\sigma > 0$ is the volatility. Consider a function

$$F(S(t), t) = S(t)e^{(r-\delta)(T-t)}$$

where T is a constant. What is the process followed by F? (That is, what is dF?) **make sure that you have no "S" values in you final SDE**.

Remark: *F(S,t) is the futures price of a futures contract maturing at time T, but the question does not require that you know this.*

***Solution*: Using Itô's formula, we have**

$$\frac{\partial F}{\partial S} = e^{(r-\delta)(T-t)}, \quad \frac{\partial^2 F}{\partial S^2} = 0 \quad \text{and} \quad \frac{\partial F}{\partial t} = -(r-\delta)e^{(r-\delta)(T-t)}S(t) = -(r-\delta)F(t).$$

Then

$$\begin{aligned}
\mathrm{d}F &= \frac{\partial F}{\partial t}\mathrm{d}t + \frac{\partial F}{\partial S}\mathrm{d}S(t) + \frac{1}{2}\frac{\partial^2 F}{\partial S^2}(\mathrm{d}S(t))^2 \\
&= -(r-\delta)F(t)\mathrm{d}t + e^{(r-\delta)(T-t)}\mathrm{d}S(t) \\
&= -(r-\delta)F(t)\mathrm{d}t + e^{(r-\delta)(T-t)}(\mu-\delta)S(t)\mathrm{d}t + e^{(r-\delta)(T-t)}\sigma S(t)\mathrm{d}X(t) \\
&= -(r-\delta)F(t)\mathrm{d}t + (\mu-\delta)F(t)\mathrm{d}t + \sigma F(t)\mathrm{d}X(t) \\
&= (\mu - r)F(t)\mathrm{d}t + \sigma F(t)\mathrm{d}X(t).
\end{aligned}$$

where the fourth equality simplifies the result by using the fact that $F(t) = e^{(r-\delta)(T-t)}S(t)$. The initial condition is $F(0) = e^{(r-\delta)(T-t)}S(0)$.

Question 3 (4 points): A practical example of using Ito's Lemma

Suppose that the prices of two (non-dividend paying) stocks and a risk-free asset follow the processes:

$$\begin{aligned}
dS_{1t} &= \mu_1 S_{1t}dt + \sigma_1 S_{1t}dX_{1t} \\
dS_{2t} &= \mu_2 S_2 dt + \sigma_2 S_{2t}dX_{2t} \\
dB_t &= rB_t dt
\end{aligned}$$

where X_{1t} and X_{2t} are correlated Brownian motions with correlation ρ. Additionally, μ_1, μ_2, σ_1, σ_2, r are positive constants.

a) Consider the process for, which is the product of the two stocks, $Y_t = S_{1t}S_{2t}$. Find dY_t. Show that it is possible to write dY_t in terms of a new Brownian motion X_{3t}.

$dY = \mu_3 Y dt + \sigma_3 Y dX_3$

where

$\mu_3 = \mu_1 + \mu_2 + \rho\sigma_1\sigma_2$
$\sigma_3 = \sqrt{(\sigma_1^2+\sigma_2^2+2\rho\sigma_1\sigma_2)}$

Now, consider a geometric average call option where the payoff is $V(S_{1T}, S_{2T}, T) = \max((S_{1T}S_{2T})^{1/2} - K, 0)$

b) Write down this payoff in terms of Y_T.

Solution: $V(Y,T) = \max(Y^{1/2} - K, 0)$

c) Determine the stochastic process followed by $Z_t = Y_t^{1/2}$.

Note you can always combine two Brownian motions to create a new one. In particular, we can write $\sigma_3 dX_{3t} = a dX_{1t} + b dX_{2t}$ where $\sigma_3 = \sqrt{(a^2+b^2+2\rho ab)}$, and you can check that $E[dX_{3t}] = 0$ and $Var[dX_{3t}] = dt$.

$dZ = \frac{1}{2}(\mu_3 - \frac{1}{4}\sigma_3^2)Zdt + \frac{1}{2}\sigma_3 Z dX_3$

d) Using your answer to c) determine the real world expected return of the geometric average of two stocks?

This is just the drift term in the expression above:_$\frac{1}{2}(\mu_3 - \frac{1}{4}\sigma_3^2)$

<u>‘Question 4 (4 points): One with exchange rates….</u>

Let B_{Pt} and B_{et} be the value of the British Pounds (GBP) and Euro(EUR)-denominated money market accounts respectively and p_t be the USD/GBP exchange rate and e_t be the USD/EUR exchange rate at time t. Dropping the t subscripts we have:

$$dB_{et} = r_e B_{et} dt$$
$$dB_{pt} = r_p B_{pt} dt$$
$$dp_t = \mu_p p_t dt + \sigma_p p_t dX_{1t}$$
$$de_t = \mu_e e_t dt + \sigma_e e_t dX_{2t}$$
$$E[dX_{1t} dX_{2t}] = \rho dt$$

where X_{1t} and X_{2t} are Brownian motions and where the correlation, ρ, the risk-free rates, r_p and r_e, the drifts, μ_p, μ_e and volatilities, σ_p and σ_e, terms are constants.

Consider an option expiring at time T giving the owner to right to **exchange** the dollar value of British pounds for Euros; this option has a payoff in USD of $1000\max(p_T - e_T, 0)$

a) Determine the stochastic processes followed by the dollar value of the British Pound Bond $\mathbf{Z_{pt}} = \mathbf{p_t B_{pt}}$ and the **Euro bond** $\mathbf{Z_{et} = e_t B_{et}}$

$$\mathbf{dZe = (\mu_e + r_e)Z_e dt + \sigma_e Z_e dX_2}$$
$$\mathbf{dZy = (\mu_y + r_y)Z_y dt + \sigma_y Z_y dX_1}$$

b) Define $Y_t = e_t/p_t$. What are the dynamics of Y_t? (That is, what is dY_t? **I would write your answer in terms of a single Brownian motion** $\mathbf{X_{3t}}$).

$$\mathbf{dY = (\mu_e - \mu_p + \sigma_p^2 - \rho\sigma_e\sigma_p)Ydt + \sigma_3 YdX_3}$$

where,

$$\mathbf{\sigma_3 = \sqrt{(\sigma_e^2 + \sigma_p^2 - 2\rho\sigma_p\sigma_e)}}$$

(2 points)

Note you can always combine two Brownian motions to create a new one. In particular, we can write $\sigma_3 dX_{3t} = adX_{1t} + bdX_{2t}$ *where* $\sigma_3 = \sqrt{(a^2+b^2+2\rho ab)}$*, and you can check that* $E[dX_{3t}] = 0$ *and* $Var[dX_{3t}] = dt$.

<u>Question 5 (3 points): Black-Scholes-Merton PDE and expected returns</u>

Suppose that the price of a dividend paying stock follows the process

$$dS(t) = (\mu - \delta)S(t)dt + \sigma S(t)dX(t),$$

the price of a risk-free bond or money-market account follows $dB(t) = rB(t)dt$, and that there is an option on the stock with price $V = V(S(t), t)$. If the stock price follows the process above, the Black-Scholes-Merton pde is,

$$\frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r - \delta)S\frac{\partial V}{\partial S} + \frac{\partial V}{\partial t} = rV$$

a) Use Ito's lemma to find the dynamics of the option price, i.e. find dV. (Your answer should be in terms of the partial derivatives $\partial V/\partial t$, $\partial V/\partial S$, and $\partial^2 V/\partial S^2$.)

Solution**: We have done this in class. We have** $\mathrm{d}S(t) = (\mu - \delta)S(t)\mathrm{d}t + \sigma S(t)\mathrm{d}X(t)$ **and** $V(t) = V(t, S(t))$**. From Itô's formula,**

$$\mathrm{d}V(t) = \frac{\partial V}{\partial t}\mathrm{d}t + \frac{\partial V}{\partial S}\mathrm{d}S(t) + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}(\mathrm{d}S(t))^2$$

or

$$\begin{aligned} dV(t) &= \frac{\partial V}{\partial t}\mathrm{d}t + \frac{\partial V}{\partial S}((\mu - \delta)S(t)\mathrm{d}t + \sigma S(t)\mathrm{d}X(t)) + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}\sigma^2 S(t)^2\mathrm{d}t \\ &= \left(\frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} + (\mu - \delta)S(t)\frac{\partial V}{\partial S} + \frac{\partial V}{\partial t}\right)\mathrm{d}t + \sigma S(t)\frac{\partial V}{\partial S}\mathrm{d}X(t). \end{aligned}$$

b) Compare the drift of V that you computed in part (a) to the left-hand side of the pde above. What is the value of μ that makes the drift of V equal to the left-hand side of the pde?

Solution**: From part (a), the drift of the option price *V* is**

$$\frac{1}{2}\sigma^2 S(t)^2\frac{\partial^2 V}{\partial S^2} + (\mu - \delta)S(t)\frac{\partial V}{\partial S} + \frac{\partial V}{\partial t}.$$

Comparing this to the left-hand side of the Black-Scholes pde above,

$$\frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} + (r - \delta)S\frac{\partial V}{\partial S} + \frac{\partial V}{\partial t},$$

so if $\mu = r$ **then the drift of the option price *V* equals the left-hand side of the pde.**

c) Setting μ equal to this value also amounts to making an assumption about the expected rate of return on the option (assume that the option value $V(t) > 0$, so the expected rate of return on the option is well defined). What is this assumption?

Hint: In part (c) it might help to divide the process $dS(t) = dS(t) = (\mu - \delta)S(t)dt + \sigma S(t)dX(t)$*, by* $S(t)$*, yielding* $dS(t)/S(t) = (\mu - \delta)dt + \sigma dX(t)$*, and then recall that the drift can be interpreted as the expected change in the process. Thus,* $\mu - \delta$ *is the expected rate of return on the stock. Similarly, the expected rate of return on the option is the drift of the option price, divided by its value* V*.*

Solution**: If we assume** $\mu = r$ **, then the stochastic process for the stock price becomes** $\mathrm{d}S(t) = (r - \delta)S(t)\mathrm{d}t + \sigma S(t)\mathrm{d}X(t)$**. Dividing both sides by *S*(*t*), we have** $\mathrm{d}S(t)/S(t) = (r - \delta)\mathrm{d}t + \sigma\mathrm{d}X(t)$**. The interpretation of this is that if** $\mu = r$ **the expected rate of return on the stock is the risk-free interest rate** $(r - \delta)$**.**
The stochastic differential equation describing the movement of the option price becomes

$$\mathrm{d}V(t) = \left(\frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (\mu - \delta)S(t)\frac{\partial V}{\partial S} + \frac{\partial V}{\partial t} \right)\mathrm{d}t + \sigma S(t)\frac{\partial V}{\partial S}\mathrm{d}X(t).$$

Dividing both sides by *V*(*t*),

$$\mathrm{d}V(t)/V(t) = \left[\left(\frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r - \delta)S(t)\frac{\partial V}{\partial S} + \frac{\partial V}{\partial t} \right) \Big/ V(t) \right]\mathrm{d}t + \left[\sigma S(t)\frac{\partial V}{\partial S}/V(t) \right]\mathrm{d}X(t).$$

Similarly, dividing both sides of the pde by *V*(*t*)

$$\left(\frac{1}{2}\sigma^2 S(t)^2 \frac{\partial^2 V}{\partial S^2} + (r - \delta)S(t)\frac{\partial V}{\partial S} + \frac{\partial V}{\partial t} \right) \Big/ V(t) = r.$$

Thus if $\mu = r$ the expected rate of return on the option price is also the risk-free rate *r*.

d) Now return to the scenario in Q2, where $\mathrm{d}S(t) = (\mu - \delta)S(t)\mathrm{d}t + \sigma S(t)\mathrm{d}X(t)$. If we set the drift of the stock equal to the drift from parts (b) and (c) above then what is the process followed by F, (i.e what is dF) now?

***Solution*: From above $\mathrm{d}F = (\mu - r)F(t)\mathrm{d}t + \sigma F(t)\mathrm{d}X(t)$. Now, μ = r, and so this becomes**

$$\begin{aligned} \mathrm{d}F &= (r - r)F(t)\mathrm{d}t + \sigma F(t)\mathrm{d}X(t) \\ &= \sigma F(t)\mathrm{d}X(t) \end{aligned}$$

i.e F is driftless under risk-neutral probabilities, or F is a martingale (see later!).